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A Review of Environmental Impact Assessment Practice in Nigeria: An Algorithmic Re-design and Sectoral Audit Using Akwa Ibom State as a Case Study

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Abstract:

Environmental Impact Assessment (EIA) is the principal regulatory instrument through which Nigeria reconciles accelerated infrastructure expansion with ecological stewardship; yet, more than three decades after the EIA Act of 1992, persistent gaps between statutory intent and field-level compliance continue to erode its effectiveness. This study undertakes a comprehensive review of the Nigerian EIA framework — its legal architecture, procedural sequence, sectoral scope, and ambient-quality standards — and proposes an algorithmically re-designed nine-stage workflow that embeds continuous feedback loops, mandatory follow-up monitoring, and digitally verifiable public-participation gates. A descriptive-analytical case study of Akwa Ibom State is conducted, in which 50 mandatory-list projects are systematically categorised across seventeen sectors and visualised. Results reveal a pronounced concentration in infrastructure and construction (18%), housing (14%), and petroleum (10%), with airports and ports the least represented (2% each). Benchmarking the conventional Nigerian process against the proposed ideal algorithm exposes structural weaknesses in screening transparency, scoping disclosure, and post-decision auditing. The study contributes (i) the first sector-disaggregated EIA inventory of Akwa Ibom State, (ii) a reproducible algorithmic template for compliance re-engineering, and (iii) a policy-actionable reform package aligning Nigeria's EIA practice with international best practice and the 2030 Sustainable Development Agenda.

Keywords: Compliance monitoring; Environmental Impact Assessment; Environmental governance; Nigeria; Public participation; Sectoral analysis; Sustainable development.

1. Introduction

Nigeria's accelerated pace of industrialisation, urban expansion, and hydrocarbon extraction has placed unprecedented pressure on its ecological systems, making a rigorously enforced Environmental Impact Assessment (EIA) regime indispensable to sustainable national development (Moriah, 2025; Nuwahereza, 2024; African Development Bank [AfDB], 2025). Globally, EIA serves as the principal anticipatory instrument for integrating environmental considerations into project planning (Kamijo, 2022). Nigeria adopted this instrument through the EIA Act of 1992 (as amended in 2004), domesticating principles set out by the World Commission on Environment and Development and operationalised by the United Nations Environment Programme (Federal Republic of Nigeria [FRN], 1992; Echefu & Akpofure, 2024).

Notwithstanding this statutory architecture, the Nigerian EIA system exhibits a recurrent and well-documented divergence between legislative intent and field-level compliance (Sam-Amobi, Ubani, Efobi, & Ajukwara, 2024; Wasah, Ogah, & Samaila, 2024). Comparative scholarship from Cameroon (Taza-Asaba, 2013), Kenya (Katumi, 2017), Rwanda (Mukama, Karume, Ndayambaje, & Iiyambo, 2025), South Africa (Jennings, 2009), and India (Islam & Wang, 2022) consistently identifies four cross-cutting pathologies: weak political will, fragmented institutional mandates, perfunctory public participation, and an almost universal failure to operationalise post-decision follow-up (Pinto, Morrison-Saunders, Bond, Pope, & Retief, 2019; Smith, 2024). These pathologies compromise the entire EIA lifecycle — from impact characterisation to monitoring and enforcement (Ekeoma, Wirnkor, Ngozi, & Ezichi, 2023).

Within Nigeria, prior reviews have addressed the historical evolution of EIA legislation (Nuwahereza, 2024), construction-sector practitioner perspectives (Kehinde, 2016), and compliance determinants in selected industrial cities (Sam-Amobi et al., 2024). What remains conspicuously absent is a sector-disaggregated, sub-national audit that simultaneously (i) inventories projects falling under the mandatory EIA study list, (ii) benchmarks the prevailing procedural workflow against an algorithmically optimised reference design, and (iii) translates the empirical findings into reform-actionable governance recommendations. The present study addresses this gap using Akwa Ibom State — a coastal Niger Delta sub-region that combines intensive hydrocarbon activity, large-scale infrastructure delivery, and ecologically sensitive mangrove and estuarine ecosystems — as a representative case (Ite, Ufot, Ite, Isaac, & Ibok, 2024).

Three research questions guide the inquiry: (RQ1) What is the sectoral distribution of EIA-mandated projects in Akwa Ibom State, and which sectors exert disproportionate regulatory pressure? (RQ2) How does the Nigerian EIA workflow compare with an algorithmically idealised process embedding feedback loops, follow-up monitoring, and digitally verifiable participation gates? (RQ3) Which institutional reforms are most likely to transform Nigerian EIA from a compliance ritual into a substantive sustainability instrument?

The remainder of the paper is organised as follows. Section 2 details the methodology, regulatory framework, ambient-quality standards, and project-selection criteria. Section 3 presents the sectoral inventory and the proposed algorithmic re-design. Section 4 discusses convergent insights from cross-jurisdictional evidence and outlines a reform package. Section 5 concludes with contributions, limitations, and research directions.

2. Methodology

This study adopts a descriptive-analytical research design that combines (a) systematic legal-doctrinal review of Nigerian EIA instruments, (b) algorithmic process modelling of the conventional and proposed nine-stage EIA workflows, and (c) a sectoral inventory of fifty (n = 50) EIA-mandated projects in Akwa Ibom State. Project records were drawn from publicly disclosed Federal Ministry of Environment registers and corroborated against AfDB (2025) and Environmental Assessment Department (EAD, 2025) sources. Inclusion was restricted to projects formally appearing on, or unambiguously satisfying, the mandatory EIA study list defined by FRN (2021).

2.1 Environmental Impact Assessment Process:

The study formalises the EIA workflow as a nine-stage decision algorithm, aligned with international best-practice guidance (Dalal-Clayton & Scott-Brown, 2024) and the procedural requirements of FRN (2021): (1) Screening, (2) Scoping, (3) Baseline Studies, (4) Impact Prediction, (5) Impact Evaluation, (6) Mitigation Design, (7) Reporting and Review, (8) Decision-Making, and (9) Follow-up Monitoring and Auditing. Each stage is treated as a node bounded by an explicit input requirement, decision rule, and verifiable output, enabling deterministic comparison with the prevailing process (Pinto et al., 2019).

2.2 Contents of an EIA Report:

An EIA report compliant with FRN (1992, as amended) must disclose: (i) project description and rationale; (ii) policy, legal, and administrative framework; (iii) baseline environmental and socio-economic conditions; (iv) significant environmental impacts; (v) analysis of alternatives, including the no-project alternative; (vi) mitigation or environmental management plan; (vii) monitoring plan; (viii) interagency and public/NGO involvement; and (ix) executive summary in non-technical language.

2.3 Regulatory Framework:

Nigeria's EIA regime is anchored on the EIA Act, Cap. E12, Laws of the Federation of Nigeria (FRN, 1992), supplemented by the EIA Procedures and Charges Regulations (FRN, 2021), the National Environmental (Air Quality Control) Regulations (FRN, 2014), and over thirty subsidiary instruments administered by the Federal Ministry of Environment and the National Environmental Standards and Regulations Enforcement Agency (NESREA). The regime distinguishes three project categories: Category I (mandatory full EIA), Category II (project-specific assessment), and Category III (presumed environmentally beneficial).

2.4 Mandatory EIA Project Categories:

FRN (2021) prescribes mandatory EIA for, inter alia: agriculture (>500 ha conversion); airports; drainage and irrigation (>5 000 ha); land reclamation (>50 ha); fisheries (>50 ha aquaculture); forestry (>500 ha conversion); housing (>50 ha or >100 units); industry (chemical, iron and steel, cement); large-scale infrastructure and construction; ports; mining and quarrying within 3 km of settlements; petroleum (production, refining, pipelines >50 km); power generation (>10 MW thermal); railways and mass-transit lines; resorts and recreational facilities; and waste-management facilities.

2.5 Ideal versus Conventional EIA Algorithm:

The conventional Nigerian workflow (EAD, 2025) follows a predominantly linear sequence terminating at the decision-making stage, with monitoring treated as discretionary. The ideal algorithm proposed in this study restructures the process as a closed-loop control system in which (a) screening outputs feed back into a digitally maintained registry of consultants and Terms of Reference; (b) scoping decisions are publicly disclosed before baseline studies commence; (c) follow-up monitoring is treated as a mandatory gate whose data outputs trigger adaptive re-mitigation when threshold exceedances occur (Pinto et al., 2019; Smith, 2024). This closed-loop topology directly addresses the enforcement deficit identified by Kamijo (2022) across thirty-seven developing-country EIA systems. A maturity-scale comparison of the two workflows is presented graphically in Figure 3.

2.6 Air-Quality Standards Framework:

Ambient air-quality limits in Nigeria are codified in FRN (2014). Table 1 reproduces the principal limits applied during EIA baseline characterisation and post-decision compliance monitoring.

Table 1: Selected Nigerian ambient air-quality limits (FRN, 2014).

Pollutant	Averaging Period	Limit Value
Sulphur dioxide (SO_2)	1 hour	350 µg/m³
Sulphur dioxide (SO_2)	24 hours	125 µg/m³
Nitrogen dioxide (NO_2)	1 hour	200 µg/m³
Carbon monoxide (CO)	8 hours	10 mg/m³
Ozone (O_3)	8 hours	100 µg/m³
PM_{10}	24 hours	150 µg/m³
$PM_{2.5}$	24 hours	75 µg/m³
Lead (Pb)	Annual mean	0.5 µg/m³

2.7 Study Area and Project Selection:

Akwa Ibom State, located in Nigeria's South-South geopolitical zone, encompasses approximately 7 081 km² of coastal, estuarine, and mangrove ecosystems and hosts the Qua Iboe Terminal, the Ibom Deep Seaport, the Ibom Industrial City, and the Lagos–Calabar Coastal Highway corridor (Ite et al., 2024). Its concentration of high-impact infrastructure within ecologically sensitive habitat makes it a critical case for EIA evaluation. A purposive sample of 50 mandatory-list projects was assembled and categorised across seventeen sectors.

3. Results

3.1 Sectoral Distribution of EIA-Mandated Projects:

The sectoral inventory (Table 2) shows that infrastructure and construction projects constitute 18.0% (n = 9) of EIA-mandated activity in Akwa Ibom State, followed by housing (14.0%; n = 7), petroleum (10.0%; n = 5), and transportation (8.0%; n = 4). Mining and quarrying combined account for 10.0% (n = 5). The lowest-represented categories are airports and ports, at 2.0% each (n = 1). The full distribution is reported in Table 2 and visualised in Figure 1, Figure 2, and Figure 4.

Table 2: Sectoral distribution of EIA-mandated projects in Akwa Ibom State (n = 50).

Sector / Project Category	Count (n = 50)	Share (%)
Agriculture	2	4.0
Airports	1	2.0
Drainage & Irrigation	2	4.0
Land Reclamation	2	4.0
Fisheries	2	4.0
Forestry	2	4.0
Housing	7	14.0
Industry	2	4.0
Infrastructure / Construction	9	18.0
Ports	1	2.0
Mining / Quarrying	3	6.0
Petroleum	5	10.0
Power Generation	2	4.0
Quarries	2	4.0
Railway	2	4.0

Sector / Project Category	Count (n = 50)	Share (%)
Transportation	4	8.0
Resorts & Recreation	2	4.0
TOTAL	**50**	**100.0**

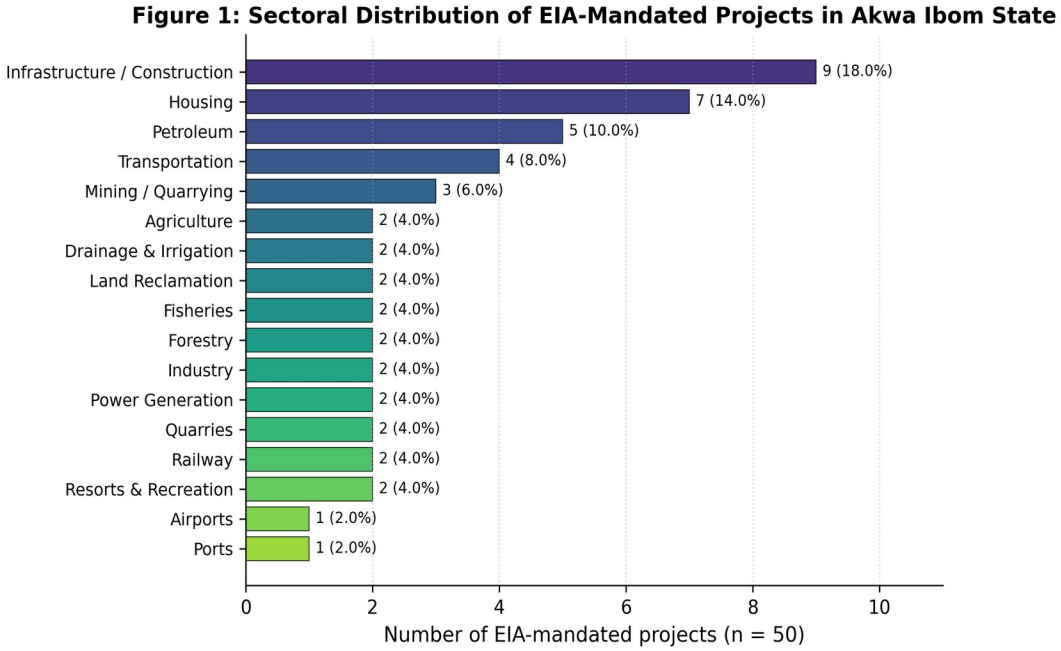

Figure 1: Sectoral distribution of EIA-mandated projects in Akwa Ibom State (n = 50). Source: Authors (2025).

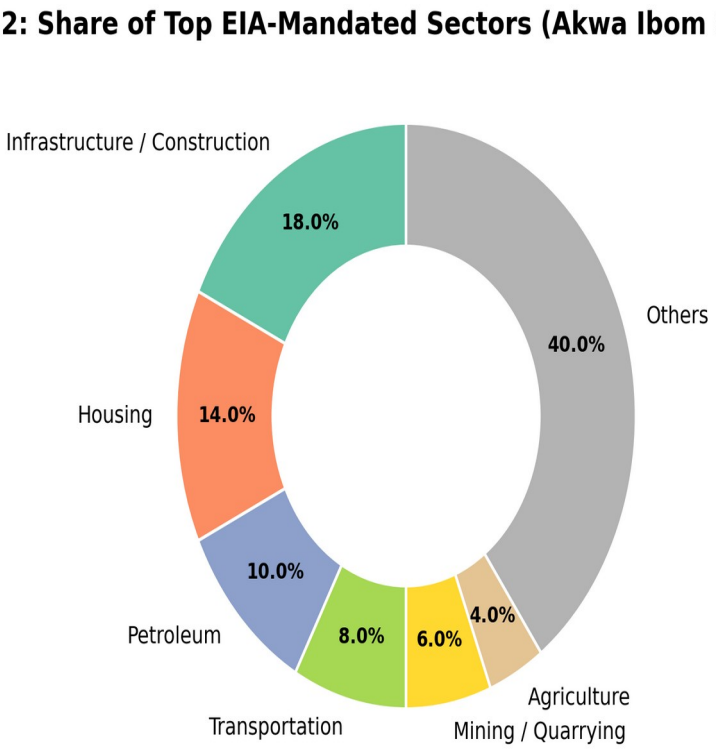

Figure 2: Share of the six largest EIA-mandated sectors in Akwa Ibom State, with all remaining sectors aggregated as "Others". Source: Authors (2025).

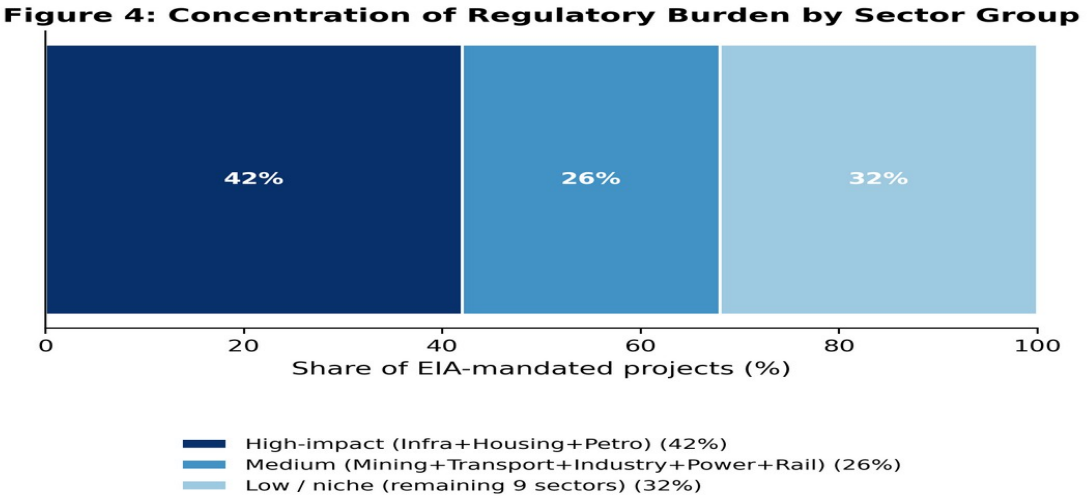

Figure 4: Concentration of regulatory burden by sector group, showing that 42% of EIA-mandated activity is generated by only three sectors (infrastructure, housing, petroleum). Source: Authors (2025).

3.2 Comparative Algorithm Audit:

Benchmarking the conventional workflow (EAD, 2025) against the proposed ideal algorithm exposes four structural deficits: (i) the absence of a publicly searchable consultant registry; (ii) non-disclosure of finalised Terms of Reference prior to baseline data collection; (iii) discretionary, rather than mandatory, follow-up monitoring; and (iv) the lack of automated triggers for adaptive re-mitigation when monitoring thresholds are exceeded. Each deficit corresponds to a specific node in the proposed closed-loop algorithm and is therefore directly remediable through targeted procedural reform. Figure 3 visualises this gap on an eight-criterion maturity scale.

Figure 3: Conventional vs Proposed Ideal EIA Algorithm (0–5 maturity scale)

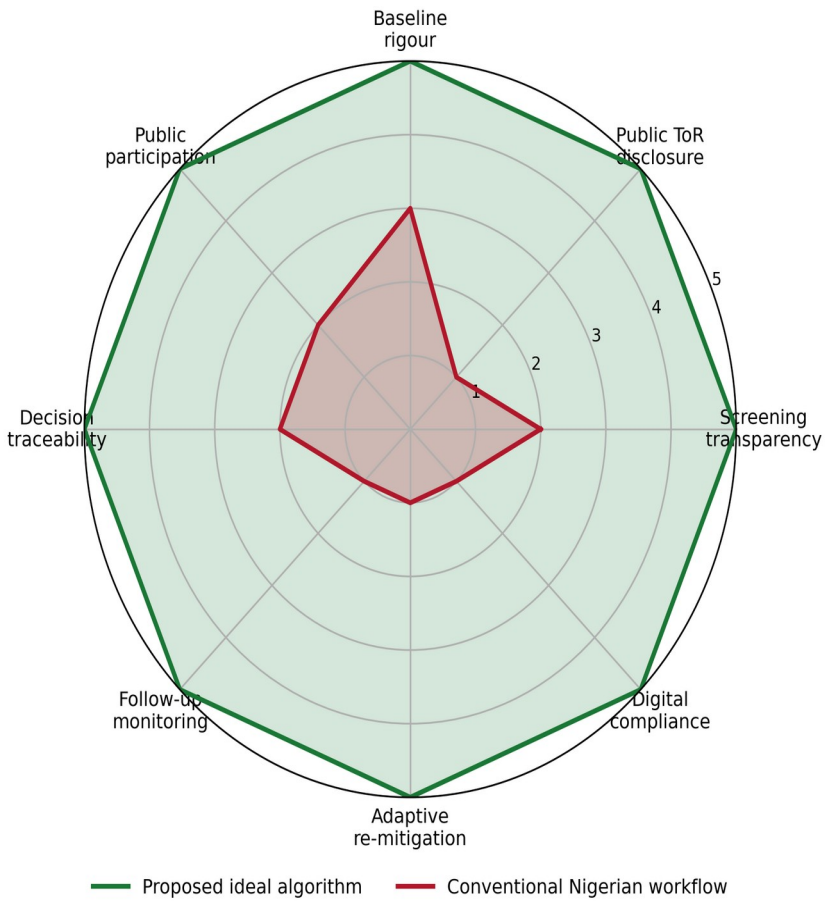

Figure 3: Maturity-scale comparison (0–5) of the conventional Nigerian EIA workflow against the proposed closed-loop algorithm across eight institutional criteria. Source: Authors (2025).

4. Discussion

The empirical concentration of EIA-mandated activity in infrastructure (18%), housing (14%), and petroleum (10%) is congruent with national patterns of capital expenditure and confirms that a small number of high-impact sectors disproportionately determine the regulatory burden. This finding converges with Wasah et al. (2024) on dam infrastructure in north-central Nigeria, with Sam-Amobi et al. (2024) on industrial compliance in Aba, and with Ite et al. (2024) on hydrocarbon-sector externalities. Taken together, these three sources suggest a sectorally asymmetric compliance challenge that no single jurisdictional study could surface — reinforcing the case for sector-tailored, rather than uniform, enforcement protocols.

The persistent weakness of public participation deserves particular scrutiny. While Section 7 of the EIA Act (FRN, 1992) formally guarantees stakeholder engagement, the operational reality reduces participation to procedural notification (Moriah, 2025; Echefu & Akpofure, 2024). For participation to function as a substantive sustainability instrument, six principles must be operationalised concurrently: it must be context-adaptive to local socio-cultural dynamics; informative and proactive in early notification; communicatively diverse across demographic strata; inclusive of marginalised constituencies; educative and consensus-oriented; and imputable, in that documented public input demonstrably influences final decisions (Dalal-Clayton & Scott-Brown, 2024).

Convergent evidence from Pinto et al. (2019), Smith (2024), and Mukama et al. (2025) further establishes that EIA effectiveness is determined less by the rigour of the predictive assessment than by the institutional quality of post-decision follow-up. This convergence motivates the closed-loop topology proposed in Section 2.5: monitoring data must be both publicly accessible and procedurally consequential, automatically re-opening the mitigation node when thresholds are exceeded.

Building on these convergences, the following institutional reforms are necessary and sufficient to align Nigerian EIA practice with international best practice: (1) statutory registration of EIA consultants together with their specific Terms of Reference; (2) mandatory disclosure of consultant identity and expertise within EIA reports; (3) publication of finalised Terms of Reference before baseline studies commence; (4) guaranteed public access to completed EIA reports prior to decision-making; (5) public disclosure of all screening decisions, approvals, and attached conditions; and (6) digitalisation of follow-up monitoring through a centralised compliance portal interoperable with NESREA enforcement workflows.

4.1 Limitations:

Three limitations bound the present study. First, the sample is restricted to publicly disclosed mandatory-list projects in a single sub-national jurisdiction; sub-threshold and informal-sector activities are necessarily excluded. Second, project categorisation relies on self-reported sectoral classifications, which may understate cross-sectoral overlap. Third, the algorithmic re-design is a normative reference model whose empirical validation will require pilot deployment across at least three states with divergent ecological and political-economic profiles.

5. Conclusion

This study has audited Nigeria's Environmental Impact Assessment regime through a sector-disaggregated case study of fifty mandatory-list projects in Akwa Ibom State and has proposed a closed-loop algorithmic re-design to remediate the structural weaknesses uncovered. Three contributions follow. First, the study delivers the first sector-disaggregated EIA inventory of Akwa Ibom State, establishing that infrastructure (18%), housing (14%), and petroleum (10%) dominate the regulatory landscape. Second, it formalises a reproducible nine-stage algorithm whose closed-loop topology embeds follow-up monitoring as a mandatory, rather than discretionary, gate. Third, it specifies a six-point reform package translating cross-jurisdictional evidence into immediately actionable governance instruments.

Beyond the academic contribution, the findings provide policymakers, NESREA enforcement officers, and EIA practitioners with a strategic framework for transforming compliance from a procedural ritual into a substantive instrument of sustainable development — a transformation indispensable if Nigeria is to align infrastructure delivery with the 2030 Sustainable Development Agenda and the IPCC AR6 mitigation pathways (IPCC, 2021).

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Conflict of Interest:

The authors declare no competing financial or personal interests that could have influenced the work reported in this paper.

Data Availability Statement:

All data supporting the findings of this study are reported within the paper. The compiled project inventory and categorisation worksheet are available from the corresponding author on reasonable request.

Author Contributions (CRediT):

Samuel Edet Bassey: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Visualization, Writing – Original Draft. Otuekong Edet Bassey: Conceptualization, Methodology, Validation, Supervision, Writing – Review & Editing, Project Administration, Corresponding Author.

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